

Simulated Port Data API

JNPA Digital Twin — Proof of Concept

API Reference · Version 2.0

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This document specifies the HTTP interface through which simulated Jawaharlal Nehru Port operational data is made available for integration testing. It covers authentication, the data group catalogue, record retrieval, file retrieval, and error handling.

Item	Value
Interface version	2.0
Protocol	HTTPS / JSON
Character encoding	UTF-8
Time zone	Asia/Kolkata (UTC+05:30)
Timestamp format	RFC 3339 with explicit offset
Authentication	Bearer token, obtained by client key exchange
Data groups	13
Records	6,625

1. Overview

The interface exposes simulated equivalents of the JNPA sample data pack. Data is retrieved in two stages. A group query returns indexed records containing metadata and a file reference; the file reference is then exchanged for the file itself through a separate endpoint.

Stage	Endpoint	Returns
1	GET /v2/groups/{group}/records	Indexed JSON. Metadata only, each record carrying a file reference.
2	GET /v2/files/{fileRef}	The file identified by the reference.

Records are returned in descending order of publication time. Responses are limited to data published at or before the instant the request is received. Each response carries an `asOf` field stating that instant.

1.1 Base URL

`https://dt.jnpa.in/poc-api-data-access/v2`

1.2 Common response fields

Field	Type	Description
<code>asOf</code>	string	Instant at which the response was evaluated.
<code>group</code>	string	Group identifier.
<code>delivery</code>	string	indexed, report or static.
<code>order</code>	string	asc or desc.
<code>count</code>	integer	Records in this response.
<code>matched</code>	integer	Records matching the filter, ignoring pagination.
<code>hasMore</code>	boolean	Further pages exist.
<code>nextCursor</code>	string	Cursor for the next page; null on the final page.
<code>items</code>	array	The records.

2. Authentication

2.1 Client key

Each organisation is issued a client key derived from its registered electronic mail address:

```
clientKey = base64( sha256(email).hexdigest()[ :20] )
```

The address is lower-cased and stripped of surrounding whitespace before hashing.

Registered address	Client key
dtinfo@jnport.gov.in	MzQ0MTMzODJiNTE1N2U2ZTY1MTY=

2.2 Token exchange

POST	/v2/auth/token
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The client key is exchanged for a bearer token. The client key is not accepted on data endpoints.

Request

```
POST /v2/auth/token
Content-Type: application/json
```

```
{ "clientKey": "MzQ0MTMzODJiNTE1N2U2ZTY1MTY=" }
```

Response

```
{
  "accessToken": "eyJzdWIiOiJjbGlfMzQ0MTMzODJiNTE1Ii4uLg",
  "tokenType": "Bearer",
  "expiresIn": 3600,
  "expiresAt": "2026-07-31T19:24:11+05:30",
  "scopes": ["groups:read", "files:read"],
  "client": { "id": "cli_34413382b515", "organisation": "JNPA" }
}
```

2.3 Presenting the token

Authorization: Bearer <accessToken>

Property	Value
Lifetime	3600 seconds
Scopes	groups:read, files:read
On expiry	HTTP 401 with reason expired; obtain a new token
Transport	HTTPS required

3. Data groups

GET

/v2/groups

Returns the catalogue of data groups with the number of records currently available in each.

#	Group identifier	Name	Delivery
1	nlp-marine	NLP Marine	indexed
2	bathymetry	Sea Channels and Bathymetry	static
3	port-craft-pilot	Port Craft and Pilot	indexed
4	shipping-lines	Shipping Lines	indexed
5	customs	Customs	indexed
6	edi-messages	EDI Messages	indexed
7	berthing-reports	Daily Berthing Reports	report
8	gate-documents	Form 13, EIR and PIN	indexed
9	rail-fois	NLDS and FOIS	indexed
10	rail-form11-icd	Form 11 and ICD Rail	indexed
11	transport	Transport Data	indexed
12	daily-reports	Performance and Daily Status Reports	report
13	cfs-ecy	CFS and Empty Yard	indexed

Delivery	Meaning
indexed	Records carry a file reference. The file is retrievable.
report	Report content returned directly as JSON. No file reference is issued.
static	Reference material distributed with the sample pack. Not served by the interface.

4. Retrieving records

GET

/v2/groups/{group}/records

4.1 Query parameters

Parameter	Type	Default	Description
from	date or instant	—	Lower bound on publication time, inclusive.
to	date or instant	—	Upper bound, inclusive. A bare date denotes the end of that day. Clamped to the present.
since	instant	—	Lower bound, exclusive. Intended for incremental retrieval.
until	instant	—	Upper bound, inclusive. Clamped to the present.
type	string	—	Message type. Accepts IGM, OOC, SMTP, FORM11, RAKE, DBR, LEO, E-DO.
order	string	desc	desc or asc.
limit	integer	50	Records per page. Permitted range 1 to 500.
cursor	string	—	Pagination cursor returned as nextCursor.
date	date	—	Report groups only. A single report date.
terminal	string	—	Report groups only. Terminal code.

4.2 Record fields

Field	Type	Description
recordId	string	Identifier for the record.
group	string	Group the record belongs to.
messageType	string	Message or report type code.
messageName	string	Type name in plain language.
publishedAt	string	Publication instant, RFC 3339.
containerCount	integer	Containers named by the message. Zero where not applicable.
vesselCall	string	Vessel call number, where the message relates to a call.
summary	string	Short description of the message content.
file.fileRef	string	Reference used to retrieve the file.
file.mediaType	string	Media type of the file.
file.sizeBytes	integer	Size of the file in bytes.
file.checksumSha256	string	SHA-256 of the file content.
file.url	string	Path at which the file is retrievable.

4.3 Example

GET /v2/groups/customs/records?type=IGM&limit=2
Authorization: Bearer <accessToken>

```
{
  "group": "customs",
  "delivery": "indexed",
  "order": "desc",
  "asOf": "2026-07-31T23:11:05+05:30",
  "count": 2,
  "matched": 102,
  "hasMore": true,
  "nextCursor": "ref_ODI43f-Z8ksV1G6",
  "items": [
    {
      "recordId": "rec_Mzk10AMFiG7dVJ4G",
      "group": "customs",
      "messageType": "CHPOI03",
      "messageName": "Import General Manifest",
      "publishedAt": "2026-07-11T01:54:00+05:30",
      "containerCount": 873,
      "vesselCall": "INNSA1NS0S0768",
      "summary": "Import General Manifest - 272 lines covering 873 containers",
      "file": {
        "fileRef": "ref_Mzk10AMFiG7dVJ4G",
        "mediaType": "application/xml",
        "sizeBytes": 793401,
        "checksumSha256": "9f1ea0e44d3a5b9e5509196a882b2e75e5dd0...",
        "url": "/v2/files/ref_Mzk10AMFiG7dVJ4G"
      }
    }
  ]
}
```

4.4 Incremental retrieval

Retain the publication time of the last record processed and supply it as since with order=asc. The bound is exclusive, so no record is returned twice and none is skipped.

GET /v2/groups/nlp-marine/records?since=2026-07-31T06:00:00%2B05:30&order=asc&limit=200

5. Retrieving files

GET /v2/files/{fileRef}

The file reference is taken from `file.fileRef` in a record. References are signed and are rejected if altered. The `files:read` scope is required.

5.1 Request

```
GET /v2/files/ref_Mzk10AMFiG7dVJ4G
Authorization: Bearer <accessToken>
If-None-Match: "9f1ea0e44d3a5b9e..."
```

5.2 Response headers

Header	Description
Content-Type	Media type of the file.
Content-Length	Size in bytes.
Content-Disposition	attachment, with the file name.
ETag	SHA-256 of the content, quoted.

5.3 Status codes

Status	Condition
200	The file is returned.
304	The supplied If-None-Match matches the current content.
401	Token absent, invalid or expired.
403	Token lacks the <code>files:read</code> scope.
404	Reference unknown, altered, or referring to a record not yet published.

5.4 Integrity

The ETag equals the `file.checksumSha256` value supplied with the record. Comparing the two confirms the file was received without modification.

6. Service endpoint

GET**/v2/health**

Liveness check. Requires no token.

```
{ "status": "ok", "uptimeSeconds": 86400 }
```

7. Errors

Errors are returned as JSON with an error code and a message. Additional fields may be present.

```
{ "error": "unauthorized", "message": "Token is not valid", "reason": "expired" }
```

Status	Code	Condition
400	bad_parameter	A parameter is malformed or outside its permitted range.
400	bad_cursor	The cursor was not issued by this interface.
400	bad_request	The request body is not valid JSON.
401	unauthorized	Token absent, invalid or expired; or client key not recognised.
403	forbidden	The token does not carry the required scope.
404	unknown_group	No group with that identifier.
404	not_found	No such endpoint, or no file for that reference.
404	invalid_reference	The file reference is malformed or has been altered.
413	payload_too_large	The request body exceeds the permitted size.
429	rate_limited	The per-minute request allowance has been exhausted.
500	internal_error	Unexpected condition.

7.1 Rate limiting

Each client has a per-minute allowance. The remaining allowance is returned in the `RateLimit-Remaining` response header. Exceeding the allowance returns 429.

7.2 Security headers

Header	Value
Strict-Transport-Security	max-age=31536000; includeSubDomains
X-Content-Type-Options	nosniff
X-Frame-Options	DENY
Referrer-Policy	no-referrer
Cache-Control	no-store

8. Endpoint summary

Method	Path	Scope	Purpose
GET	/	—	Service description.
GET	/v2/health	—	Liveness check.
POST	/v2/auth/token	—	Exchange the client key for a bearer token.
GET	/v2/groups	groups:read	Data group catalogue.
GET	/v2/groups/{group}/records	groups:read	Indexed records for a group.
GET	/v2/files/{fileRef}	files:read	Retrieve a file by reference.

8.1 Worked sequence

```
# 1. obtain a token
curl -X POST https://dt.jnpa.in/poc-api-data-access/v2/auth/token \
  -H 'Content-Type: application/json' \
  -d '{"clientKey":"MzQ0MTMzODJiNTE1N2U2ZTY1MTY="}'

# 2. list the groups
curl -H "Authorization: Bearer $TOKEN" https://dt.jnpa.in/poc-api-data-access/v2/groups

# 3. retrieve records
curl -H "Authorization: Bearer $TOKEN" \
  'https://dt.jnpa.in/poc-api-data-access/v2/groups/customs/records?type=IGM&limit=25'

# 4. retrieve a file
curl -H "Authorization: Bearer $TOKEN" -OJ \
  https://dt.jnpa.in/poc-api-data-access/v2/files/ref_Mzk10AMFiG7dVJ4G
```